

Guide to the Output of the Final-Segment Spectral Analysis Code (Version 11)

1 What the code does, and what it does not do

The code takes a sequence of pairs (n, μ_n) , where μ_n is the minimum modulus among the eigenvalues of the matrix attached to index n , and asks whether $\{\mu_n\}$ oscillates with a fixed period once n is large.

Two design decisions distinguish version 11 from its predecessors.

First, **only the final segment is analyzed**. The code locates the points where the data changes character (for instance, where an initial linear ramp gives way to a plateau), and restricts all analysis to the last of the resulting segments, on the premise that the oscillation phenomenon emerges only for large n . Earlier segments are located and displayed for context but are not analyzed.

Second, **the code does not render a verdict**. It does not label the data “periodic” or “not periodic.” It reports a collection of numerical metrics, each with its score, and leaves the judgment to you. This guide explains precisely what each metric measures and how to read it, so that the interpretation is yours to make on an informed basis.

The remainder of the guide is organized as follows: Section 2 develops, from scratch, the small amount of Fourier analysis and statistics the metrics rest on; Section 3 is a glossary of every abbreviation that appears in the report; Sections 4–7 walk through the report block by block; Section 8 describes the figure; and Section 9 discusses how to weigh the metrics together.

2 Mathematical background

Everything below is elementary, but because the intended reader has not met these particular constructions, they are defined in full.

2.1 The discrete Fourier transform and the power spectrum

Let $x_0, x_1, \dots, x_{n-1}$ be a finite real sequence. Its *discrete Fourier transform* (DFT) is the sequence of complex numbers

$$c_k = \sum_{j=0}^{n-1} x_j e^{-2\pi i j k / n}, \quad k = 0, 1, \dots, n-1.$$

Each c_k records “how much” of the complex exponential of frequency $f_k = k/n$ (in cycles per index step) is present in the sequence. The quantity

$$P_k = |c_k|^2$$

is called the *power* at frequency f_k , and the function $k \mapsto P_k$ is the *power spectrum*. A sequence that is close to a pure sinusoid of period T has its power concentrated near the frequency $f = 1/T$;

the period corresponding to a frequency f is simply $T = 1/f$. The *Fast Fourier Transform* (FFT) is an algorithm computing the DFT in $O(n \log n)$ arithmetic operations; it produces exactly the c_k above, only faster. Nothing in the analysis depends on the speed—“FFT” may be read throughout as “the coefficients c_k .”

2.2 Windowing

The DFT implicitly treats the finite sample as one period of an n -periodic sequence. If x_0 and x_{n-1} differ substantially, this imagined wrap-around introduces a spurious jump, whose energy smears across all frequencies and contaminates the spectrum—an effect called *spectral leakage*. To suppress it one multiplies the data, before transforming, by a *window* that tapers smoothly to zero at both ends. The code uses the *Hann window*

$$w_j = \frac{1}{2} \left(1 - \cos \frac{2\pi j}{n-1} \right), \quad j = 0, \dots, n-1,$$

and divides the resulting power by $\frac{1}{n} \sum_j w_j^2$ to compensate for the energy the taper removes. The only consequence you need to retain: the windowing is a standard hygiene step, and it slightly attenuates content at the very highest representable frequency (period 2).

2.3 The autocorrelation function

Given the same sequence with mean $\bar{x}$, its *autocorrelation function* (ACF) at integer lag $k \geq 0$ is

$$\rho(k) = \frac{\sum_{j=0}^{n-1-k} (x_j - \bar{x})(x_{j+k} - \bar{x})}{\sum_{j=0}^{n-1} (x_j - \bar{x})^2}.$$

It is the correlation of the sequence with itself shifted by k places, and $\rho(0) = 1$. If the sequence repeats with period T , then x_j and x_{j+T} rise and fall together, so $\rho(T)$ is close to 1; thus a sharp local maximum of ρ at lag $k = T$ is direct evidence of repetition with period T . The ACF is, in a sense, complementary to the power spectrum: the spectrum decomposes the sequence into frequencies, while the ACF measures self-similarity under translation. For data built from sharp, non-sinusoidal repetitions the ACF is the more reliable period detector, because a non-sinusoidal shape scatters its Fourier power across many frequencies but still repeats cleanly under the right translation.

2.4 Why a “null model” is needed

A finite stretch of *any* sequence—even one with no genuine periodicity—will show some bumps in its power spectrum and some nonzero autocorrelations, purely by accident. To decide whether an observed bump is larger than such accidents, one needs a reference: a model of “structureless” data against which the observation is compared. This reference is the *null model*.

The code’s null model is the *first-order autoregressive process*, written AR(1):

$$X_n = \rho X_{n-1} + \varepsilon_n,$$

where the ε_n are independent draws from a normal (Gaussian) distribution and $\rho \in (-1, 1)$ is a fixed parameter, the lag-1 autocorrelation. This process has short-term memory—each value is pulled

toward a multiple of its predecessor—but no periodicity whatever. It is the standard stand-in for “correlated but aperiodic” data. Its expected power spectrum is known in closed form,

$$S(f) = \frac{\sigma^2(1 - \rho^2)}{1 - 2\rho \cos(2\pi f) + \rho^2},$$

(due to Gilman, Fuglister, and Mitchell, 1963), and the code uses this curve as the baseline against which spectral peaks are measured. When $\rho > 0$ the baseline slopes upward toward low frequencies—historically called *red noise*, by analogy with red light occupying the low-frequency end of the visible spectrum.

2.5 Thresholds, the chi-squared distribution, and the Bonferroni correction

Under the null model, the power at a single frequency, divided by the baseline value there, behaves like a standard random quantity whose distribution is known: one-half of a *chi-squared* variate with two degrees of freedom. (The chi-squared distribution with d degrees of freedom is the law of $Z_1^2 + \dots + Z_d^2$ for independent standard normal Z_i ; only the resulting numerical thresholds matter here.) Fixing a small probability α , one can compute a level L such that, under the null, the power exceeds L at a given frequency with probability only α . Taking $\alpha = 0.05$ and $\alpha = 0.01$ gives the 95% and 99% thresholds reported in the table.

One subtlety is essential. The spectrum is examined at many frequencies at once—say m of them. Even if each individually has only a 5% chance of a false exceedance, across m independent frequencies one expects about $0.05m$ false exceedances. The *Bonferroni correction* compensates by testing each frequency at the stricter level α/m instead of α , so that the chance of *any* false exceedance across the whole spectrum returns to α . The reported thresholds already incorporate this correction; m is the number of frequency bins examined (those with period at most $n/3$).

2.6 The white-noise band for the autocorrelation

There is an analogous reference for the ACF. If the data were pure independent noise, then for a fixed lag the sample autocorrelation $\rho(k)$ would be approximately normal with mean 0 and standard deviation $1/\sqrt{n}$. Hence an autocorrelation larger in magnitude than about $1.96/\sqrt{n}$ would arise by chance only about 5% of the time: this is the *white-noise 95% band* (a consequence of Bartlett’s formula; “white noise” denotes a sequence of independent, identically distributed values, with a flat power spectrum, by analogy with white light). Reporting an ACF strength as a multiple of this band tells you how many “noise standard deviations” the observed self-similarity stands above what pure noise would supply.

2.7 Percentiles and the Spearman correlation

Two descriptive statistics appear in the envelope diagnostics. The q -th *percentile* of a finite set of numbers is the value below which a fraction $q/100$ of the data lies; the code uses the 10th and 90th percentiles ($p10$, $p90$) to summarize the spread of the amplitude envelope while ignoring outliers. The *Spearman rank correlation* between two sequences is the ordinary (Pearson) correlation computed after replacing each value by its rank; it ranges in $[-1, 1]$ and measures monotone association—whether one quantity tends to increase with the other—without assuming the relationship is linear. A Spearman correlation near ± 1 means strictly monotone; near 0 means no monotone trend.

3 Glossary of abbreviations

ACF	Autocorrelation function; see §2.
ACF strength	The value $\rho(T)$ of the ACF at a detected period T ; a number in $[-1, 1]$, larger meaning cleaner repetition.
ACF/band	The ACF strength divided by the white-noise 95% band $1.96/\sqrt{n}$; how many noise standard deviations the value exceeds.
AR(1)	First-order autoregressive process, the aperiodic null model.
Bonferroni	The correction dividing the per-frequency significance level by the number of frequencies tested.
FFT	Fast Fourier transform; computes the Fourier coefficients c_k .
Hann window	The smooth taper applied before the FFT to limit spectral leakage.
MA(w)	Moving average of width w : each value replaced by the mean of the w values centered on it. Used to smooth the data before recomputing the ACF, so that finer structure is averaged away and longer periods can surface.
n	The number of points in the final segment (the analyzed data).
$n/3$	The longest period considered; with fewer than three full cycles in the data, a putative period cannot be told from a trend.
p10, p90	The 10th and 90th percentiles of the amplitude envelope.
Power	$ c_k ^2$, the squared magnitude of a Fourier coefficient; printed in scientific notation, e.g. $1.44\text{e}+05 = 1.44 \times 10^5$.
Power/95%	The ratio of a peak's power to the 95% threshold at the same frequency; greater than 1 means the peak exceeds that threshold.
rho (ρ)	The lag-1 autocorrelation of the analyzed signal, i.e. the fitted AR(1) parameter; not to be confused with $\rho(k)$, the ACF at lag k .
Spearman	Rank correlation; here, between the amplitude envelope and position, measuring how monotone the envelope is.
std dev	Standard deviation, the root-mean-square deviation from the mean.
thr	Threshold (the 95% or 99% Bonferroni-corrected level).
white-noise 95% band	$\pm 1.96/\sqrt{n}$; the range within which a sample autocorrelation is consistent with pure independent noise.

4 The segmentation block

The report first prints the data range, the number of points, and the range of the moduli. It then prints the *segmentation*.

A *breakpoint* is an index at which the data’s overall level changes abruptly; the stretch between consecutive breakpoints is a *segment*. Breakpoints are found by a standard changepoint algorithm (the Pelt method from the `ruptures` library), followed by a cleanup step that merges adjacent segments of nearly equal slope, so that a single straight ramp—which the underlying algorithm tends to chop into several pieces—is recombined into one. Each segment is printed with its index range, length, and value range; the final segment is marked `<-- ANALYZED`. Only that segment is carried into the metrics below.

A caveat carried over from earlier versions: this style of changepoint detection cannot, in general, distinguish a slow oscillation spanning only a few cycles from a sequence of genuine level changes. If the segmentation has visibly chopped an evident oscillation into equal pieces, treat the segmentation as unreliable for that input and consider analyzing the data as a single segment.

5 The detrending and envelope blocks

Detrending

Within the final segment a straight line is fitted by least squares and subtracted, leaving the *residual*. This removes any constant slope so that the spectral analysis sees only the fluctuation about the trend. Three numbers are reported:

linear-fit slope The slope of the fitted line.

residual std dev

The standard deviation of the residual; the typical size of the fluctuation about the line.

residual / value scale

The residual standard deviation divided by the largest absolute value in the segment. When this ratio is extremely small (below 10^{-8}) the segment is, to numerical precision, an exact straight line, and any apparent oscillation is floating-point noise; the code says so and stops. Otherwise it is a rough measure of how large the fluctuations are relative to the signal.

Envelope

The *amplitude envelope* is a smooth curve tracing the size of the fluctuations: the absolute residual, smoothed. A genuine oscillation whose amplitude decays or grows (think of $\sin(2\pi n/T)/n$) has an envelope that moves monotonically; a flat band of noise has an envelope that merely wanders. Three numbers are reported:

dynamic range (p90/p10)

The ratio of the envelope’s 90th to its 10th percentile; how much the amplitude varies across the segment.

monotonicity (|Spearman|)

The absolute Spearman correlation between the envelope and position; how steadily, as opposed to erratically, the amplitude trends.

normalization applied

Whether the code divided the residual by the envelope. It does so only when the dynamic range exceeds 3 *and* the monotonicity exceeds 0.6—that is, only for a genuine, steady amplitude trend. This guards against “whitening” a band of noise, which can otherwise manufacture a spurious period-2 signal. When normalization is applied, the power figures that follow describe the normalized signal, and should be read as evidence about *the existence* of a period, not about physical amplitude.

6 The spectral baseline and FFT-peak blocks

The *spectral baseline* block reports the fitted AR(1) parameter **rho**, the number of *frequency bins tested* (the m of the Bonferroni correction), and the *max credible period*, equal to $n/3$.

The *FFT peaks* block lists the local maxima of the power spectrum whose period is at most $n/3$, strongest first. For each it gives the period, the power, the Bonferroni-corrected 95% and 99% thresholds at that frequency, and the ratio **Power/95%**. Read the ratio as the headline number: a value above 1 means the peak rises above the 95% threshold (it would occur by chance, under the null, with probability below the corrected level); a value far above 1—in the test runs, a genuine oscillation produced ratios in the tens—is strong spectral evidence for a period, while the many peaks with ratios well below 1 are the ordinary ripples of noise and should be disregarded. No line is drawn for you; the ratios are the evidence.

7 The multi-scale ACF block

To find the period by self-similarity rather than by frequency content, the code computes the ACF not only of the signal but of several moving-average-smoothed copies of it (the **raw** signal, then **MA(3)**, **MA(5)**, and so on). Smoothing averages away short-scale structure, letting longer periods reveal themselves; the first ACF peak at each scale is recorded.

The block first prints the *white-noise 95% band*, $\pm 1.96/\sqrt{n}$, the magnitude an autocorrelation would need merely to be noteworthy against pure noise. It then tabulates each detected period with the scale at which it was found, its *ACF strength*, and the ratio **ACF/band**. One period is labeled *fundamental*—the one whose integer multiples account for the most of the others (an oscillation of period T produces ACF peaks at $T, 2T, 3T, \dots$, and the code reports T as fundamental and the rest as *integer multiples*). As with the FFT, read **ACF/band** as the headline: it states how many noise standard deviations the self-similarity exceeds. A fundamental with a large ACF strength *and* a large **ACF/band** is strong, frequency-independent evidence for a period.

8 The figure

A single four-panel figure is produced, all for the final segment:

1. **Data with linear trend.** The segment (blue) and the fitted line (red); confirm the line captures the slope without tracking the wiggles.
2. **Analysis signal.** What the FFT and ACF actually receive (residual, possibly envelope-normalized); it should fluctuate about zero.

3. **Power spectrum.** Power against frequency on a logarithmic vertical axis, with the AR(1) baseline (black) and the 95% (green) and 99% (red) thresholds; peaks rising above the dashed curves are the candidates.
4. **Power vs period.** The same information against period rather than frequency, restricted to periods between 2 and $n/3$, which is usually the easier axis to read.

9 Reading the metrics together

Because the code issues no verdict, the weighing is yours. The two independent lines of evidence are the spectral one (**Power/95%** in the FFT block) and the correlational one (**ACF/band**, with the ACF strength, in the ACF block). They have complementary strengths: the FFT is decisive for clean sinusoids, the ACF for sharp or non-sinusoidal repetition. Concordance between them—a strong FFT peak at period T and a strong ACF fundamental at the same T —is the most persuasive evidence. Discordance deserves scrutiny: a strong ACF peak at period 2 with no FFT support is expected, since the Hann window suppresses the highest frequency; a strong FFT peak with a weak ACF, in a short segment, may be a single chance exceedance among many bins.

Three structural caveats bear on every reading. The thresholds rest on the AR(1) null model and the assumptions behind it; they are calibrations, not theorems about your data. The $n/3$ cap hides any period longer than a third of the final segment, so a genuinely long period can be invisible. And the ACF strength has no associated significance test beyond the white-noise band, which assumes independence the data may not have; the **ACF/band** ratio is therefore a guide, not a proof. Used together, and with the figure in view, the metrics are intended to let you reach your own conclusion on a transparent basis.